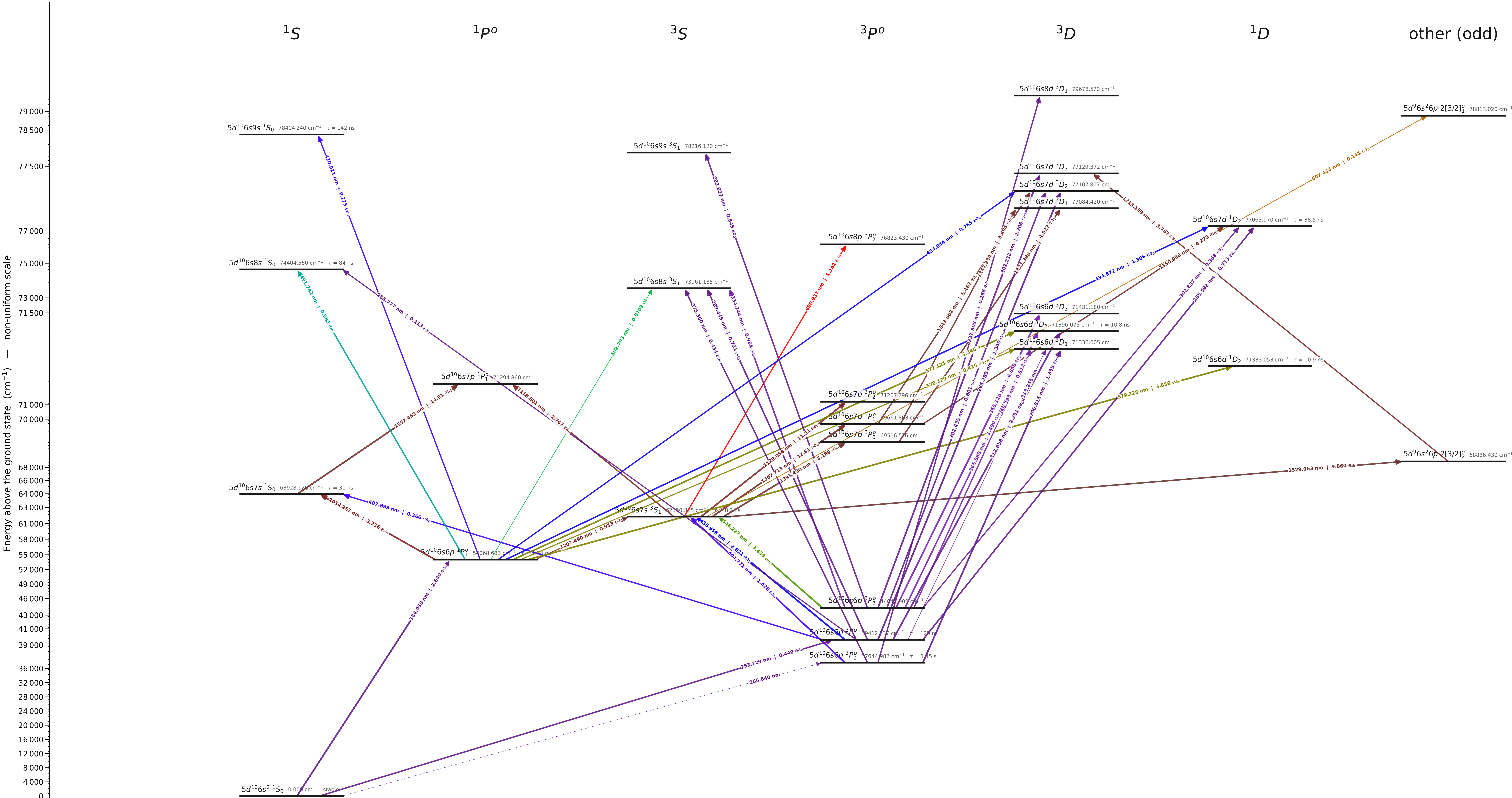
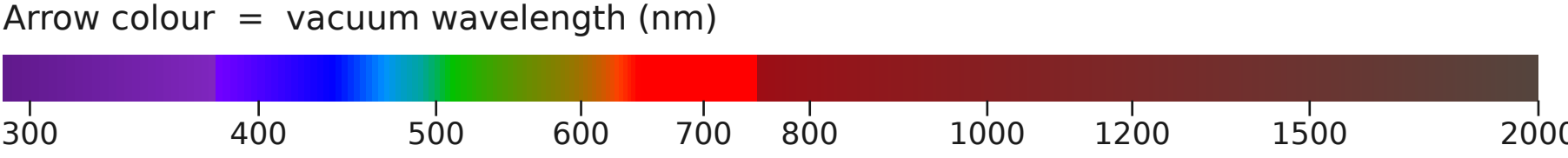


Mercury

Hg (Hg I). Strongest classified transitions below 2 μm (NIST): 27 levels, 47 lines.

Hg ⁺ ionisation limit 84184.15 cm⁻¹ = 10.43750 eV (118.787 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|(U||er||J')|$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ (U er J') $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ¹P ₁ ^o	184.9499	–	1620.9383	2.640	7.440e+08	118.8	99.7	184.9 nm resonance line
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₁ ^o	253.7291	253.6528	1181.545677	0.440	7.992e+06	1.272	100	253.7 nm intercombination line: MOT / cooling, optical pu...
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₀ ^o	265.6397	265.5606	1128.5682	–	6.920e-01	1.101e-07	100	265.6 nm optical lattice clock transition
5d ¹⁰ 6s6p ³P ₀ ^o – 5d ¹⁰ 6s8s ¹S ₀	285.7770	285.6930	1049.0434	0.113	1.100e+06	1.895	9.24	
5d ¹⁰ 6s6p ³P ₀ ^o – 5d ¹⁰ 6s6d ³D ₁	296.8150	296.7283	1010.0315	1.335	4.600e+07	–	–	296.7 nm
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s6d ³D ₂	312.6579	312.5673	958.8513	2.231	6.000e+07	14.74	71.3	312.7 nm
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s6d ³D ₁	313.2462	313.1555	957.0505	–	–	–	–	313.2 nm, optical pumping out of the 253.7 nm MOT
5d ¹⁰ 6s6p ³P ₂ ^o – 5d ¹⁰ 6s6d ³D ₃	365.1198	365.0158	821.0797	4.658	1.290e+08	–	–	365.0 nm
5d ¹⁰ 6s6p ³P ₀ ^o – 5d ¹⁰ 6s7s ³S ₁	404.7707	404.6564	740.6476	1.426	2.070e+07	19.41	17	404.7 nm
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s7s ¹S ₀	407.8988	407.7837	734.9677	0.366	4.000e+06	5.134	12.4	407.8 nm
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s7s ³S ₁	435.9561	435.8336	687.6666	2.621	5.600e+07	19.41	45.9	435.8 nm
5d ¹⁰ 6s6p ¹P ₁ ^o – 5d ¹⁰ 6s8s ¹S ₀	491.7418	491.6045	609.6543	0.583	5.800e+06	1.895	48.7	491.6 nm
5d ¹⁰ 6s6p ³P ₂ ^o – 5d ¹⁰ 6s7s ³S ₁	546.2267	546.0750	548.8425	3.439	4.900e+07	19.41	40.2	546.1 nm
5d ¹⁰ 6s6p ¹P ₁ ^o – 5d ¹⁰ 6s7s ¹S ₀	1014.2567	1013.9787	295.5785	3.736	2.710e+07	5.134	84	1014 nm
5d ¹⁰ 6s6p ¹P ₁ ^o – 5d ¹⁰ 6s7s ³S ₁	1207.4900	1207.1596	248.2774	0.913	3.200e+05	19.41	0.26	1207 nm intercombination branch of 7s 3S1

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

201Hg hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5d ¹⁰ 6s6p ³P ₁ ^o	-5454.569(3)	–	F=1/2: +13636.42 F=3/2: +5454.57 F=5/2: -8181.85
5d ¹⁰ 6s6p ³P ₂ ^o	-3352.0292(8)	–	F=1/2: +15084.13 F=3/2: +10056.09 F=5/2: +1676.02 F=7/2: -10056.09
5d ¹⁰ 6s6p ¹P ₁ ^o	1316.000	–	F=1/2: -3290.00 F=3/2: -1316.00 F=5/2: +1974.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₁ ^o	253.729	196-198	4179.95(3)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₁ ^o	253.729	200-198	-4805.52(4)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₁ ^o	253.729	202-198	-10101.224(22)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₁ ^o	253.729	204-198	-15311.60(4)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₁ ^o	253.729	199-198	-652(6)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₁ ^o	253.729	201-198	-6410(6)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₀ ^o	265.640	199-201	5729.668(8)
5d ¹⁰ 6s ² ¹S ₀ – 5d ¹⁰ 6s6p ³P ₀ ^o	265.640	198-199	654.48
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s6d ³D ₂	312.658	196-198	-289.4(12)
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s6d ³D ₂	312.658	200-198	328.0(6)
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s6d ³D ₂	312.658	202-198	686.7(6)
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s6d ³D ₂	312.658	204-198	1038.3(6)
5d ¹⁰ 6s6p ³P ₁ ^o – 5d ¹⁰ 6s6d ³D ₁	313.246	196-198	-307.5(10)
5d ¹⁰ 6s6p ³P ₂ ^o – 5d ¹⁰ 6s6d ³D ₁	313.246	200-198	347.2(6)
5d ¹⁰ 6s6p ³P ₂ ^o – 5d ¹⁰ 6s6d ³D ₁	313.246	202-198	725.5(6)
5d ¹⁰ 6s6p ³P ₂ ^o – 5d ¹⁰ 6s6d ³D ₁	313.246	204-198	1097.8(6)

Shift = ν(first isotope) – ν(second isotope).